# FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – II

## JEE (Main)-2023

### TEST DATE: 20-01-2023

## ANSWERS, HINTS & SOLUTIONS

## *Physics*

**PART – A**

**SECTION – A**

1. C

Sol. For the sector, $y_{cm} = \frac{4R}{3}\frac{\sin(\phi/2)}{\phi} = \frac{2R}{\pi} \quad \left(\because \phi = \frac{\pi}{3}\right)$

For the arc, $y_{cm} = \frac{2R\sin(\phi/2)}{\phi} = \frac{3R}{\pi}$

2. A

Sol. If acceleration of block 2 is $a_2$ to the right, then

$5a_2 = a_r + a = 6 + 4$

$\boxed{a_2 = 2m/s^2}$

3. C

Sol. $\alpha = \frac{\tau}{I} = \frac{FL}{\frac{mL^2}{3}} = \frac{3F}{m\ell} \quad \Rightarrow \quad a_{cm} = \frac{\ell}{2}\alpha = \frac{3F}{2m}$

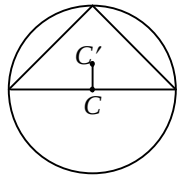
$\Sigma F_x = ma_{cm}$

$N_x + F = \frac{3F}{2} \quad \Rightarrow N_x = \frac{F}{2}$

$N_y = Mg$

4. C

Sol. C.M. of triangle is at $\frac{a}{3}$ from centre of circular plate




$$m_{\text{triangle}} = \frac{m_{\text{circle}}}{\pi}$$

$$y_{CM} = \frac{M_y - m_y}{M - m}$$

On solving, $\frac{a}{3(\pi - 1)}$

5. C

Sol. $-\Delta U = \Delta k_e$

$$-\left[U_B - U_A\right] = \frac{1}{2}mv^2$$

Ball will leave when normal at B = 0

$$\therefore\ mg\sin\beta - N_B = \frac{mv_B^2}{R} \Rightarrow 0 \quad \text{at B} \quad \Rightarrow\ mg\sin\beta = \frac{mv_B^2}{R}$$

6. D

Sol. Block number 1 and 2*n* will suffer one collision each, block number 2 and 2*n* – 1 will suffer two collision each similarly block number *n* and *n* + 1 will suffer *n* collision each. So total number of collisions will be 2(1 + 2 + … + *n*) = *n* (*n* + 1)

7. A

Sol. $r = R\left(\frac{\ell_1}{\ell_2} - 1\right) = 132.4\left(\frac{70}{60} - 1\right) \approx 22.1\,\Omega$

8. B

Sol. 100 Ω, 25 Ω and 20 Ω are in parallel.

Their, net resistance is 10 Ω

$\therefore$ $R_{net}$ = 4 Ω + 10Ω + 6Ω = 20 Ω

V = i $R_{net}$ = 80 V

9. A

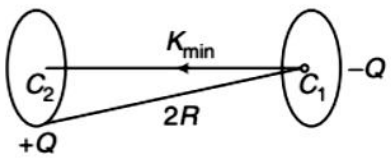
Sol. Current decreases $\frac{20}{30}$ times or $\frac{2}{3}$ times. Therefore, net resistance should become $\frac{2}{3}$ times.

$$\therefore\ R + 50 = \frac{3}{2}(2950 + 50)$$

Solving we get, R = 4450 Ω

10. A

Sol.




$$k_{C_1} + U_{C_1} = k_{C_2} + U_{C_2}$$

$$k_{min} + qV_{C_1} = 0 + qV_{C_2} \qquad \ldots(i)$$

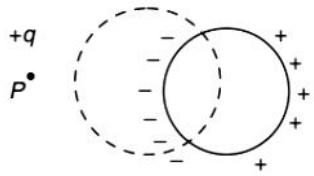
$$V_{C_1} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{2R} - \frac{Q}{R}\right)$$

$$V_{C_2} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{2} - \frac{Q}{2R}\right)$$

Substituting these values in Eq. (i), we can find $K_{min}$.

11. C

Sol.




The induced charges on conducting sphere due to +q charge at P are as shown in figure.
Now, net charge inside the closed dotted surface is negative. Hence, according to Gauss's theorem net flux is negative.

12. A

Sol. $p^2V$ = constant

$$\therefore \left(\frac{nRT}{V}\right)^2 V = \text{constant}$$

$$\therefore \quad T^2 \propto V$$

or $\quad T \propto \sqrt{V}$

V is made three times. So, T will becomes $\sqrt{3}$ times.

13. A

Sol. $W_{net} = 2p_0V_0$

$Q_{+ve} = Q_{ABC} = W_{ABC} + \Delta U_{ABC}$

= area under the graph + $nC_v\Delta T$

$$= (3p_0V_0) + n\left(\frac{3}{2}R\right)(T_C - T_A)$$

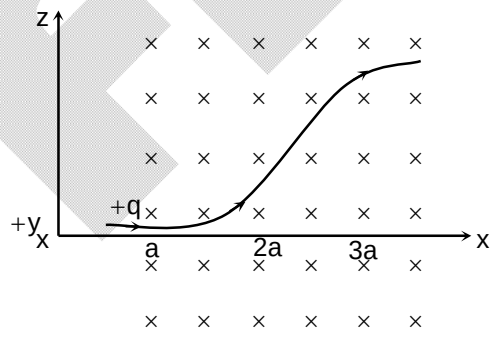
$$= (3p_0V_0) + \frac{3}{2}(p_CV_C - p_AV_A)$$

$= 10.5\ p_0V_0$

$$\eta = \frac{W_{net}}{Q_{+ve}} = \frac{4}{21}$$

14. A

Sol.




15. C

Sol. Conceptual

16. A

Sol. $\tan\phi = \dfrac{X_C}{R} \qquad \dfrac{1}{\omega C} = R$

$\tan\dfrac{\pi}{4} = \dfrac{X_C}{R} \qquad \dfrac{1}{\omega R} = C$

$X_C = R$

i leads emf $\therefore$ circuit is LC

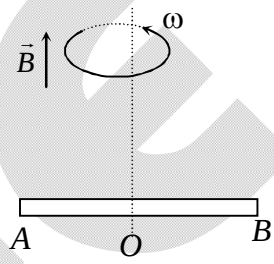
17. D

Sol. Potential difference between

*O* and *A* is $V_A - V_O = \dfrac{1}{2}Bl^2\omega$

*O* and *B* is $V_B - V_O = \dfrac{1}{2}Bl^2\omega$

So $V_A - V_B = 0$




18. B

Sol. $\lambda = \dfrac{h}{\sqrt{2mE}} \qquad \Rightarrow \qquad \lambda \propto \dfrac{1}{\sqrt{E}}$

$\Rightarrow \qquad \dfrac{\lambda_1}{\lambda_2} = \sqrt{\dfrac{E_2}{E_1}} \qquad \Rightarrow \qquad \dfrac{10^{-10}}{0.5\times10^{-10}} = \sqrt{\dfrac{E_2}{E_1}}$

$\Rightarrow \qquad E_2 = 4E_1$

Hence added energy = $E_2 - E_1 = 3E_1$

19. A

Sol. Using Moseley's law, we get

$\dfrac{\lambda_2}{\lambda_1} = \dfrac{(Z_1 - a)^2}{(Z_2 - a)^2}$

$\lambda_2 = \dfrac{200\times(74-1)^2}{(78-1)^2} = 179.76\ \text{Å}$

20. C

Sol. ${}^{x}\lambda = \dfrac{\ln 2}{t_{1/2}}$

${}^{y}\lambda = \dfrac{\ln 2}{t_{1/2}}$

$\dfrac{{}^{x}A}{{}^{y}A} = \dfrac{{}^{x}\lambda\,{}^{x}N_0 e^{-{}^{x}\lambda t}}{{}^{y}\lambda\,{}^{y}N_0 e^{-{}^{2}\lambda t}}$

$e^{\ln 2} = 2$

## SECTION – B

21. 2

Sol. $D_1$ is reverse biased therefore it will act like an open circuit.

$i = \dfrac{12}{6} = 2.00A$

22. 3

Sol. Psueodo force and friction will make a force couple. Normal reaction and Mg will make a force couple.

When block is about the topple normal reaction will shift to edge.

23. 2

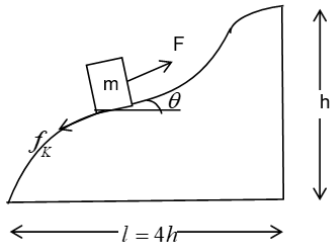
Sol. $\Delta W_f = -\int \mu mg\cos\theta ds$

$= -\mu mg\int_0^l dx$

$= -\mu mgl = 0.5mg\,4h = -2mgh$

$\Delta W_g = -mgh$

$\therefore \Delta W_{fr} / \Delta W_g = 2$




24. 5

Sol. $mg = kx$ (for $m$ to leave the ground)

$m_0 gx = \frac{1}{2}kx^2$ or $m_0 = \frac{m}{2}$ = 5 kg

25. 4

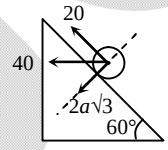
Sol. Conservation of momentum along horizontal

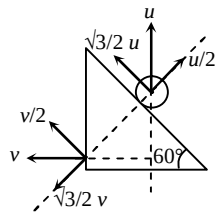
$m40 = 6\times v$ …(i)

Newton's law of restitution

$\frac{u}{2} + \frac{\sqrt{3}}{2}v = 20\sqrt{3}$ …(ii)

Linear momentum of ball will remain conserved along inclined




Before Collision

$\frac{\sqrt{3}}{2}u = 20$

$u = \frac{40}{\sqrt{3}}$ …(iii)

From (i), (ii) and (iii)

$m$ = 4 kg

26. 4

Sol. $P_{avg} = \frac{\varepsilon_{10} i_0}{2} = \frac{80}{2} = 40$

$P_{avg} = \frac{\text{Energy}}{\text{Time}}$

27. 40

Sol. Here 3rd maxima is shifted by $3\times10^{-4}$ m. It indicates fringe width increases by $1\times10^{-4}$ m.

Hence $\beta = \frac{\lambda(D+0.5)}{d} = \frac{\lambda D}{d} + 1\times10^{-4}$

or $\frac{0.5\lambda}{d} = 1\times10^{-4}$ m or $\lambda = \frac{2\times10^{-3}\times1\times10^{-4}}{0.5} = 4\times10^{-7}$ m = 40 nm

28. 30

Sol. $\lambda_{C_1} = \frac{hc}{eV} = \frac{12375}{10\times10^3} A° = 1.2375\ A°$ … (1)

$\lambda_{C_2} = 0.61875\ A°$ … (2)

$$\frac{1}{\lambda_{K_\alpha}} = (Z-1)^2 \left\{\frac{1}{1} - \frac{1}{4}\right\}\times10^7$$ … (3)

It is given

$3 \times (\lambda_{k\alpha} - 1.2375) = \lambda_{k\alpha} - 0.61875$

$\therefore$ $\lambda_{k\alpha} = 1.54338\ A°$ … (4)

Putting this value in (3)

$Z - 1 = 29$

$Z = 30$

29. 6

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$$v' = \frac{\upsilon\, v}{\upsilon - \upsilon_s}$$ …(1)

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener $v'' = \frac{(\upsilon + \upsilon_L)v'}{\upsilon}$

using (1)

$$v'' = \frac{\upsilon + \upsilon_L}{\upsilon} \times \frac{\upsilon v}{\upsilon - \upsilon_s} = \frac{(\upsilon + \upsilon_L)v}{\upsilon - \upsilon_s}$$

$$= \frac{(330+5)\times200}{330-5} = \frac{335}{325}\times200$$

$v'' = 206\,\text{Hz}$

$\therefore$ Beat frequency = $(v'' - v)$

$= 206 - 200 = 6\,\text{Hz}$

30. 3

Sol. Total heat product $= \frac{1}{2}CV^2$

$= \frac{1}{2}(2\mu F)(5)^2$

$= 25\ \mu J$

Now, this should distribute in inverse ratio of resistors, as they are in parallel.

$\therefore$ $\frac{H_{5\Omega}}{H_R} = \frac{R}{5}$

or $H_{5\Omega} = \left(\frac{R}{R+5}\right)$ (Total heat)

or $10 = \left(\frac{R}{R+5}\right)(25)$

Solving this equation, we get

$R = \left(\frac{10}{3}\right)\Omega$

# Chemistry

## PART – B

### SECTION – A

31. C

Sol. Intensity of incident radiations depends on no. of photons in the incident light and simultaneously no. of photoelectrons increases and photoelectric current increases

32. A

Sol. In all the three options groups are arranged at ortho position. So after that the value of dipole moment will depend on the electro negativity difference.

33. C

Sol. For helium is a very light gas for which attraction is ignored and do not have vanderwall constant 'a' in the equation of compressibility factor and for helium Z > 1.

34. A

Sol. $Na_2CO_3 \longrightarrow NaHCO_3$ x

$NaHCO_3 \longrightarrow H_2CO_3$ x

$NaHCO_3 \longrightarrow H_2CO_3$ y

With phenolphthalein as indication 10 mL $\times$ 1N $\times$ 2

2 mill g. equation = x

with methyl orange as indicator

10mL $\times$ 0.2N $\times$ 4 milli gram equation = x + y

y = 2 mili gram eq.

Normality = $\frac{2\text{mill gram eq.}}{20\,\text{mL}} = 0.1\text{N}$

N = nf $\times$ molarity M = 0.05 M

35. D

Sol. For first order reaction, $t = \frac{2.303}{k}\log\frac{a}{a-x}$ $t_{99.9} = \frac{2.303}{k}\log\frac{100}{100-99.9} = \frac{2.303}{k}\times 3$

$t_{50} = \frac{2.303}{k}\log\frac{2.303}{k}\log 2 = \frac{2.303}{k}\times 0.301$ $t_{99.9/t50} = 10\,\text{approx.}$

36. C

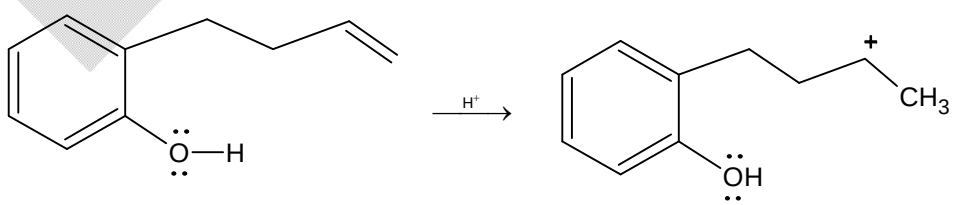
Sol. The cations will be liberated in the sequence of decreasing reduction potentials. Cations having $E^{\circ}$value < –0.83 (reduction potential of water) will not be liberated from aqueous solutions

37. C

Sol. Iodine is a bulky group due to ortho effect mesomeric effect stops at ortho position & hence a and b will be longer than C bond

38. B

Sol.




More stable carbocation forms at this position

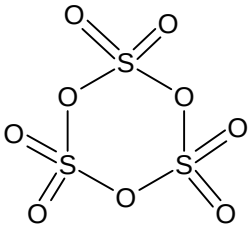
39. B

Sol. Benzilic acid involves transformation of $\alpha$–diketones to $\alpha$–hydroxyl acid by means of $OH^-$

40. C

Sol. $CO_2$, in $CO_2$ carbon is already in it's maximum oxidation state and cannot be oxidised further. While N, Cl, S are not in their maximum oxidation state.

41. D

Sol. Structure  no S – S bonds

42. B

Sol. $[Co(NH_3)_2Cl_4]^-$ = cis and trans isomerism
$AuCl_2Br_2^-$ square planar cis and trans isomersim
$[Co(NO_2)\ [NH_3]^{2+}$ = No isomerism

43. C

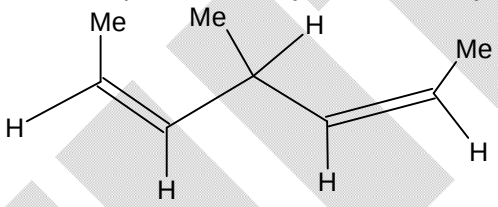
Sol. VBT is based on measurement of magnetic moment measurement. Based on that structure is being predicted.

44. B

Sol. Fehlings solution can separate reducing and non-reducing sugars.
Glucose – Reducing sugar
Sucrose – non-reducing sugar.

45. B

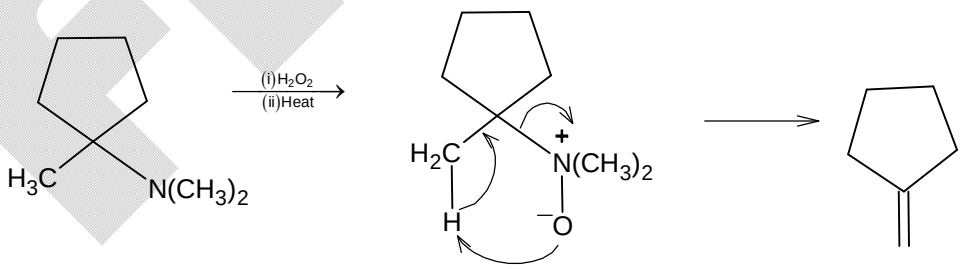
Sol. Poisoned palladius catalyst reduces Alkyne to cis-Alkene.



Optically inactive product

46. D

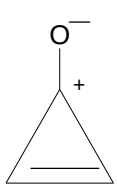
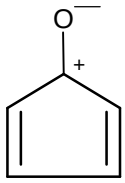
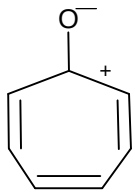
Sol.



Less stable alkene is the major product

47. B

Sol.

Aromatic Anti-aromatic Aromatic

Anti-aromatic is the least stable

48. A

Sol. Partial pressure of $O_2 = \frac{2}{7} \times 2660$ mm

Thus 1 L of $O_2$ is present at 0ºC and 760 mm

So no. of $O_2$ molecules = $\frac{0.02 \times 10^{23}}{22.4}$

PV = nRT

No. of moles of $O_2 = \frac{1\text{atm} \times 1\text{L}}{0.0821^{-1}\text{atm}} \times 273\text{k} = \frac{1}{22.4}$

49. C

Sol. They have different bond connectivity. Hence constitutional isomers.

50. C

Sol. Zone refining is based on the principal that impurities are more soluble in molten state than solid state.

Hence molten zone contains more impurities than the original metal.

## SECTION – B

51. 25

Sol.

$$AB_2 \rightleftharpoons AB(g) + B(g)$$

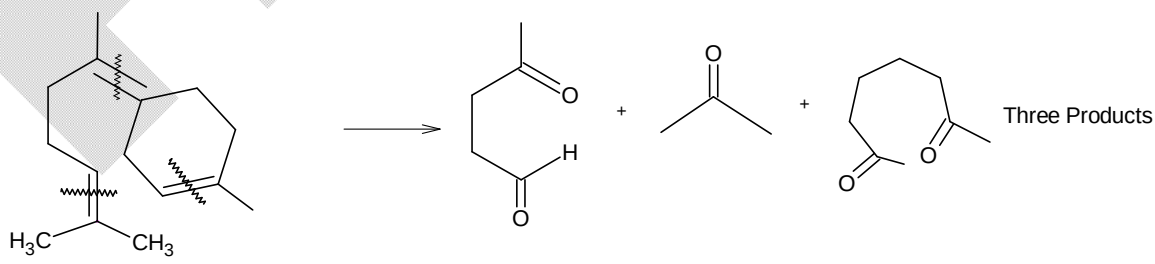
| | | |
|---|---|---|
| 500 | – | – |
| 500 – x | x | x |

Require = 600 torr = 500 – x + x + x, x = 100 torr

$K_p = \frac{P_{AB} \cdot P_B}{P_{AB_2}} = \frac{100 \times 100}{400}$ = 25 torr

52. 3

Sol.




Three Products

53. 6470

Sol. $H_2SO_4 \rightleftharpoons H^+ + HSO_4$

x – –

x – y y + Z y – Z

$H_2SO_4^- \rightleftharpoons H^+ + SO_1^{a-}$

y – Z Z + y Z

$K_{G_1} = \frac{(y-Z)(y+Z)}{x-y}$ $k_q$ $\frac{Z(Z+y)}{y-Z}$

$k_{a_1} = \infty$, $\therefore x = y$

$1.2 \times 10^{-2} = \frac{2 \times 10^{-2}}{y-Z}$

1.2 (y) – 1.22 = Z

1.2 y = 2.2Z

Y = 1.833Z

Y + z = 0.01

2.8332 = 0.01

Z = 0.0025

Y = 0.0064705 = x

$\therefore$ 6470.546

54. 2

Sol. $d_{z^2}$ orbital has no mola plane.

$Fe(CO)_5 = dsp^2 = dz^2$

$MnO_4^- = d^3sw = d_{xy}$ , $d_{yz}$, $d_{x2}$

$Co(C_2O_4)_3^{3-} = sp^3d^2$ $d_{x^2-y^2}$ and $d_{z^2}$

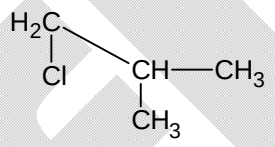
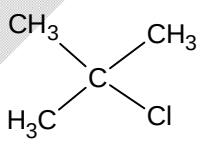
$NiCl_2(PPh_3)_2 = dsp^2 = d_{x^2-y^2}$

$PtCl_4^{2} = dsp^2 = d_{x^2-y^2}$

$Ni(CN)_4^{2} = dps^2 = d_{x^2-y^2}$

$Ni(CN)_4^{4-} = sp^3$

55. 2

Sol.  &


2 products

56. 5

Sol. $Fe_x(CO)_y \xrightarrow{\Delta} \frac{x}{2}Fe_2O_3 + yCO_2$

$Fe_x(CO)_y + \left(\frac{3x}{4} + y\right)O_2 \longrightarrow \frac{x}{2}Fe_2O_3 + y\ CO_2$

0.0026 mole

1-50 L at 25ºC, P = 44.9 mm Hg

PV = nRT

$$\frac{44.9}{760} \times 1.50L = n \times 0.0821 \times 298$$

$$= n = 0.036 \text{ mole}$$

$0.142 = 56\ x + 0.0036 \times 28$

$$x = \frac{0.0412}{56} = 0.00073$$

$$\frac{y}{x} = 4.9 \simeq 5$$

57. 182

Sol. Fcc like structure has edge length

$2r_+ + 2r_- = 520$ ppm

From radius ration

$$\frac{r^+}{r^-} = 0.414$$

$2\ (0.414\ r^-) + 2r^- = 520$ pm

$2.85\ r^- = 520$

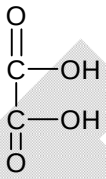
$r^- = 182$ pm

58. 84

Sol. Given wt. of complex = 3.2 gm

Molecular wt. of complex = 4 M + 48 + 264

= 4 M + 528 + 216

Molar mass of oxalic acid is  is 90 gm

$$\frac{3.2g}{4M+528} \times = \frac{1g}{90}$$

M = 84 g/ mol

59. 2

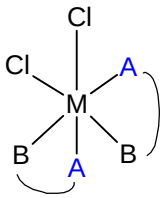
Sol. Rate = $\underbrace{\left[4.7\times10^{-5}\right][R-x]\left[OH^-\right]}_{\downarrow S_N2} + \underbrace{0.24\times10^{-5}\left[RX\right]}_{\downarrow S_N1}$

$$\%\ \text{of } SN^2 = \frac{4.7\times10^{-5}\left[RX\right]\times10^{-3}M}{0.24\times10^{-5}\left[RX\right]+4.7\times10^{-5}\left[RX\right]}\times100$$

on solving answer = 2

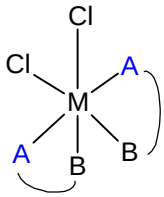
60. 8

Sol. $\left[Pt(gly)_2 Cl_2\right] = \left[Pt(AB)_2 Cl_2\right]$

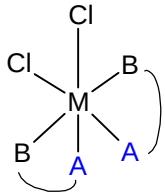



complete cis

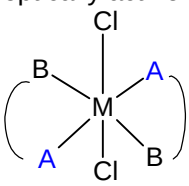
optically active = 2




optically active = 2




optically actice = 2




optically inactice
optically inactive

& Cis – $[co\ (en)_2\ Cl_2]$ is optically active isomers = 2 while trans – $[Co(en)Cl_2]$ is

# Mathematics

## PART – C

### SECTION – A

61. D

Sol. $\sin^{-1}\dfrac{|2\times3+(-1)\times6+2\times(-2)|}{\sqrt{2^2+(-1)^2+2^2}\cdot\sqrt{3^2+6^2+(-2)^2}}$

62. C

Sol. $\dfrac{dy}{dx}=\cos^{-1}x^4x^2$

$= -\cos^{-1}x^2$

$= \dfrac{2\pi}{3}\times\dfrac{1}{2^{1/4}}-\dfrac{\pi}{4}$

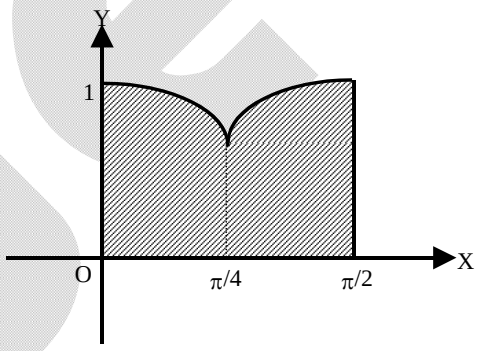
63. A

Sol. f(x) = cosx for $0 \le x \le \pi/4$

= sinx for $\pi/4 < x \le \pi/2$

Required = $2\int_0^{\pi/4}\cos x\,dx = 2\sin x\,\Big|_0^{\pi/4}$

$= \sqrt{2}$ sq. units.




64. D

Sol. $(1-x^4)\,(1+{}^9C_1x+{}^9C_2x^2+{}^9C_3x^3\ldots\ldots\ldots{}^9C_9x^9)$

Coefficient of $x^7 = {}^9C_7-{}^9C_3 = -48$

65. C

Sol. The chord of contact of tangents from $(\alpha, \beta)$ is

$\alpha x+\beta y=1$ ..... (1)

Also, $(\alpha, \beta)$ lies on $2x+y=4$, so that $2\alpha+\beta=4$

$\Rightarrow \dfrac{\alpha}{2}+\dfrac{\beta}{4}=1$

Hence, (1) passes through $\left(\dfrac{1}{2},\ \dfrac{1}{4}\right)$.

66. C

Sol. Equation of line is $(2x+3y+4)+\lambda(6x-3y+12)=0$ ...(1)

(1) is normal to circle, therefore (1) passes through (2, 0)

$\Rightarrow \lambda = \dfrac{-1}{3}$ and required equation is y = 0.

67. A

Sol. $\Rightarrow 2y\dfrac{dy}{dx}x=-y^2-\sin 2x$ $\Rightarrow y^2+2yx\dfrac{dy}{dx}=-\sin 2x$

$\Rightarrow \frac{d}{dx}(xy^2) = \frac{d}{dx}(\cos^2 x)$ $\Rightarrow xy^2 = \cos^2 x + c$

68. D

Sol. $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$

Operating $C_1 \to C_1 - C_2$

$$\begin{vmatrix} 4a^x a^{-x} & (a^x - a^{-x})^2 & 1 \\ 4b^y b^{-y} & (b^y - b^{-y})^2 & 1 \\ 4c^z c^{-z} & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & (a^x - a^{-x})^2 & 1 \\ 1 & (b^y - b^{-y})^2 & 1 \\ 1 & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 0 .$$

69. C

Sol. $0 < \frac{3}{x^2+1} \le 3$, $\frac{3}{x^2+1} = 2 \Rightarrow x = \frac{1}{\sqrt{2}}$ and $\frac{3}{x^2+1} = 1 \Rightarrow x = \sqrt{2}$

$$I = \int_0^{1/\sqrt{2}} 2dx + \int_{1/\sqrt{2}}^{\sqrt{2}} 1.dx + \int_{\sqrt{2}}^{\infty} 0\,dx = \sqrt{2} + \sqrt{2} - \frac{1}{\sqrt{2}} = 2\sqrt{2} - \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}} .$$

70. C

Sol. $\int \frac{\ln x}{(1+\ln x)^2} dx$

Let $\ln x = t \Rightarrow dx = e^t dt$

$$\int \frac{te^t}{(1+t)^2} dt = \int \frac{e^t(t+1-1)}{(1+t)^2} dt$$

$$= \int \frac{e^t(t+1-1)}{(1+t)^2} dt$$

$$= \int e^t \left\{ \frac{1}{t+1} - \frac{1}{(1+t)^2} \right\} dt$$

$$= \frac{e^t}{t+1} + c \qquad \text{since } \int e^x (f(x) + f'(x)) dx = e^x f(x) + c$$

$$= \frac{e^{\ln x}}{1+\ln x} + c = \frac{x}{1+\ln x} + c .$$

71. A

Sol. $f(x) = \log_{1/2}\log_4\log_3[(x-4)^2] \Rightarrow \log_4(\log_3[(x-4)^2]) > 0 \Rightarrow \log_3[(x-4)^2] > 1$

$\Rightarrow [(x-4)^2] \ge 4 \Rightarrow (x-4)^2 - 4 \ge 0 \Rightarrow (x-2)(x-6) \ge 0$

$\Rightarrow x \in (-\infty, 2] \cup [6, \infty)$ .

72. A

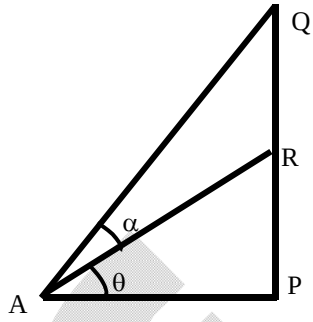
Sol. $\tan(\theta + \alpha) = \frac{PQ}{AP} = \frac{1}{n}$

$\tan\theta = \frac{PR}{AP} = \frac{(1/2)PQ}{AP} = \frac{1}{2n}$

$\alpha = (\theta + \alpha) - \theta$

$\Rightarrow \tan\alpha = \frac{\tan(\theta+\alpha) - \tan\theta}{1 + \tan(\theta+\alpha)\tan\theta}$

$= \frac{\frac{1}{n} - \frac{1}{2n}}{1 + \frac{1}{n}.\frac{1}{2n}} = \frac{n}{2n^2+1}.$




73. C

Sol. Normal to hyperbola will be

$ax\cos\theta + by\cot\theta = a^2 + b^2$

If this is same as $lx + my + n = 0$, comparing the coefficients,

$\frac{a\cos\theta}{l} = \frac{b\cot\theta}{m} = \frac{a^2+b^2}{-n}$

$\sec\theta = -\frac{an}{l(a^2+b^2)}$, $\tan\theta = \frac{-bn}{m(a^2+b^2)}$

$\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2+b^2)^2}{n^2}.$

74. A

Sol. Any point on the parabola is $(x , x^2+ 7x + 2)$

Its distance from the line $y = 3x - 3$ is given by

$P = \left|\frac{3x - (x^2+7x+2) - 3}{\sqrt{9+1}}\right| = \left|\frac{x^2+4x+5}{\sqrt{10}}\right| = \frac{x^2+4x+5}{\sqrt{10}}$ (as $x^2 + 4x + 5 > 0$ for all $x \in R$)

$\frac{dP}{dx} = 0 \Rightarrow x = -2$

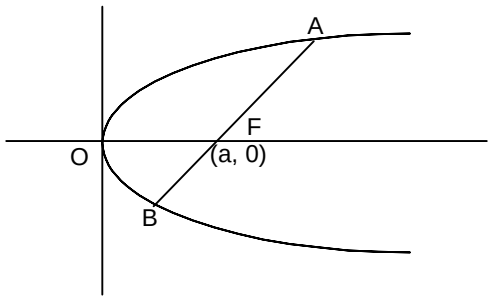
The required point $\equiv (-2, -8)$.

75. A

Sol. FA = 4, FB = 5

We know that $\frac{1}{a} = \frac{1}{AF} + \frac{1}{FB}$

$\Rightarrow a = \frac{20}{9} \Rightarrow 4a = \frac{80}{9}.$



76. D

Sol. $\frac{n}{4n^4+1}=\frac{1}{4}\left[\frac{1}{2n^2-2n+1}-\frac{1}{2n^2+2n+1}\right]$

Hence $\sum_{n=1}^{k}\frac{n}{4n^4+1}=\frac{1}{4}\left[1-\frac{1}{2k^2+2k+1}\right]=\frac{1}{4}$ as $k\to\infty$

77. C

Sol. $S-S_n<\frac{1}{300}$

$\Rightarrow \frac{1}{1-\frac{1}{3}}-\frac{1-\frac{1}{3^n}}{1-\frac{1}{3}}<\frac{1}{300} \Rightarrow \frac{3}{2}\left[1-1+\frac{1}{3^n}\right]<\frac{1}{300}$

$\Rightarrow \frac{1}{3^n}<\frac{1}{300}\times\frac{2}{3} \Rightarrow \frac{1}{3^n}<\frac{1}{450} \Rightarrow 3^n>450$

$\therefore$ least value of n = 6

78. B

Sol. Let $f(x)=x^3-3x+a$

$f'(x)=3x^2-3$.

For three distinct real roots (i) $f'(x)=0$ should have two distinct real roots $\alpha$ and $\beta$ and (ii) $f(\alpha)\,f(\beta)<0$

Here $\alpha=1$, $\beta=-1$.

Now $f(\alpha)\,f(\beta)<0$

$\Rightarrow (1-3+a)(-1+3+a)<0 \Rightarrow (a-2)(a+2)<0$

$\Rightarrow -2<a<2$.

79. C

Sol. The line $y=\sqrt{3}x$ can be written as $x=\frac{r}{2}$, $y=\frac{r\sqrt{3}}{2}$.

If this line cuts the given curve, then

$\frac{r^4}{16}+\frac{ar^3\sqrt{3}}{8}+\frac{br^2\sqrt{3}}{4}+\frac{cr}{2}+\frac{dr\sqrt{3}}{2}+6=0$.

Therefore OA. OB.OC.OD = $|r_1|\,|r_2|\,|r_3|\,|r_4| = |r_1\,r_2\,r_3\,r_4| = 96$

80. C

Sol. Plane passing through (2, 2, 1) is $a(x-2)+b(y-2)+c(z-1)=0$

It passes through (9, 3, 6)

$\Rightarrow 7a+b+5c=0$ .... (1)

It is $\perp$ to $2x+6y+6z-1=0$

$\Rightarrow 2a+6b+6c=0$ .... (2)

(1) and (2) $\Rightarrow \frac{a}{-24}=\frac{b}{-32}=\frac{c}{40}$

$\Rightarrow \frac{a}{3}=\frac{b}{4}=\frac{c}{-5}$

$\Rightarrow$ The required plane is

$3(x-2)+4(y-2)-5(z-1)=0$

$\Rightarrow 3x+4y-5z-9=0$ .

## SECTION – B

81. 9

Sol. $S.D = \dfrac{\begin{vmatrix} 10 & 2 & 3 \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}}{\sqrt{8^2+8^2+4^2}} = \dfrac{108}{12} = 9$

82. 0

Sol. Let $f(x) = x^3 + 2x^2 + 5x + 2\cos x$

$\Rightarrow f'(x) = 3x^2 + 4x + 5 - 2\sin x$

$= 3\left(x+\dfrac{2}{3}\right)^2 + \dfrac{11}{3} - 2\sin x$

Now $\dfrac{11}{3} - 2\sin x > 0 \ \forall\ x$ (as $-1 \le \sin x \le 1$)

$\Rightarrow f'(x) > 0 \ \forall\ x \Rightarrow f(x)$ is an increasing function.

Now $f(0) = 2$

$\Rightarrow f(x) = 0$ has no solution in $[0, 2\pi]$.

83. 5

Sol. $T_2 = {}^nC_1\, ab^{n-1} = 135$ .....(1)

$T_3 = {}^nC_2\, a^2 b^{n-2} = 30$ .....(2)

$T_4 = {}^nC_3\, a^3 b^{n-3} = 10/3$ .....(3)

Dividing (1) by (2)

$$\frac{{}^nC_1 ab^{n-1}}{{}^nC_2 a^2 b^{n-2}} = \frac{135}{30}$$

$$\frac{n}{\frac{n}{2}(n-1)}\frac{b}{a} = \frac{9}{2}$$

$$\frac{b}{a} = \frac{9}{4}(n-1) \quad \text{.... (5)}$$

Dividing (2) by (3)

$$\frac{\frac{n(n-1)}{2}}{\frac{n(n-1)(n-2)}{3.2}}\cdot\frac{b}{a} = 9$$

$$\therefore \frac{b}{a} = 3(n-2) \quad \text{....(6)}$$

Eliminating a and b from (5) and (6)

$\Rightarrow n = 5$

84. 4

Sol. $k = |z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ $\Rightarrow k = 4$

85. 8

Sol. $\frac{x^2}{4}+\frac{y^2}{3}=1 \Rightarrow$ a = 2, b = $\sqrt{3}$ and e = $\frac{1}{2}$

Sum of distances = $\frac{2a}{e} = \frac{4}{1/2}=8$.

86. 120

Sol. g(x) = g(x – 1 + 1) = g(x – 1) g(1)
= g(x – 2) $[g(1)]^2$ = …. = $[g(1)]^x = 3^x$

$\Rightarrow \sum_{x=1}^{4} 3^x = 3^1 + 3^2 + 3^3 + 3^4 = 120$

87. 0

Sol. $\lim\limits_{x\to 0}\left[f(x)+\log\left(1-\frac{1}{e^{f(x)}}\right)-\log(f(x))\right]$

$= \lim\limits_{x\to 0}\left[\log\left(e^{f(x)}\right)+\log\left(\frac{1-\frac{1}{e^{f(x)}}}{f(x)}\right)\right]$

$\Rightarrow \lim\limits_{x\to 0}\log\left(\frac{e^{f(x)}-1}{f(x)}\right)=0 \Rightarrow \lim\limits_{x\to 0}\left(\frac{e^{f(x)}-1}{f(x)}\right)=1$

$\Rightarrow \lim\limits_{x\to 0} f(x)=0 \Rightarrow f(0) = 0$ (as f(x) is a continuous function) .

88. 2100

Sol. First, 6 distinct digits can be selected in $^{10}C_6$ ways. Now the position of smallest digit in them is fixed i.e. position 4. Of the remaining 5 digits, two digits can be selected in $^5C_2$ ways. These two digits can be placed to the right of $4^{th}$ position in one way only. The remaining three digits to the left of $4^{th}$ position are in the required order automatically.
So n(S) = $^{10}C_6 \times {}^5C_2 = 210 \times 10 = 2100$.

89. 20

Sol. Required number of ways = $\frac{6!}{3!3!}=\frac{720}{6\times 6}=20$

90. 2

Sol. $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ = ad – bc = 0 $\Rightarrow$ whether ad = 1, bc = 1 or ad = – 1, bc = – 1

which occur in eight ways. Total number of 2 × 2 determinants from {–1, 1} is 16. Thus required probability is $\frac{8}{16}=\frac{1}{2}$.